B.E. Semester – VII
Choice Based Credit Grading Scheme with Holistic & Multidisciplinary Education (CBCGS-HME 2023)
Proposed TCET Autonomy Scheme (w.e.f. A.Y. 2024-25)

BE Information Technology								B.E (SEM: VII)		
Course Name: Web Analytics and Information Retrieval								Course Code: : PEC-IT 7023		
Teaching Scheme (Program Specific)								Examination Scheme (Academic)		
Modes of Teaching / Learning / Weightage								Modes of Continuous Assessment / Evaluation		
Hours Per Week					Theory (100)			Practical/ Oral/ Presentation (25)	Term Work (25)	Total
Theory	Tutorial	Practical	Contact Hours	Credits	ISE	IE	ESE	OR	TW	150
3	-	2@	5	4	20	20	60	25	25	
ISE: In-Semester Examination - Paper Duration – 1 Hour										
IE: Innovative Examination										
ESE: End Semester Examination - Paper Duration - 2 Hours										
Total weightage of marks for continuous evaluation of Term work/Report: Formative (40%), Timely Completion of Practical (40%) and Attendance /Learning Attitude (20%).										
Prerequisite: : Basic knowledge of programming, statistics, databases, web technologies, and data analytics.										

Course Objective: This course introduces students to the foundational concepts and applications of Web Analytics and Information Retrieval. It emphasizes the collection, analysis, and interpretation of web data, SEO strategies, and the implementation of information retrieval models to enhance digital presence and user experience.

Course Outcomes: **Upon completion of the course students will be able to:**

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy	PO Mapping	PSO Mapping
1	Explain the fundamental concepts of web analytics and information retrieval systems.	L1, L2	1, 2	1
2	Apply appropriate web analytics tools and interpret collected data effectively.	L1, L2, L3	1,2, 3, 5, 11	1, 2
3	Analyze user behaviour and SEO performance using key metrics and techniques.	L1, L2, L3, L4	2, 3, 5	1, 2
4	Evaluate different information retrieval models and implement ranking algorithms.	L1, L2, L3, L4	1, 2, 3, 5	1, 2
5	Design and execute optimization strategies using A/B testing, personalization, and recommendation systems.	L1, L2, L3, L4, L5	3, 4, 5, 11	2, 3

6	Develop real-time analytics dashboards using big data frameworks while addressing ethical considerations.	L1,L2,L3,L4,L5,L6	3, 5, 6, 10	2, 3

Detailed Syllabus:

Module No.	Topics	Hrs.	Cognitive levels of attainment as per Bloom's Taxonomy
1	**Introduction to Web Analytics and Information Retrieval** Overview of Web Analytics: metrics, KPIs, user behaviour, Basics of Information Retrieval: indexing, querying, ranking, Web data sources: server logs, APIs, clickstreams, Components of IR systems: document representation, retrieval models, Applications in IT: search engines, digital marketing, user behaviour analysis	6	L1,L2
2	**Web Analytics Techniques and Tools** Web metrics: page views, bounce rate, conversion rate, dwell time, Data collection: server logs, cookies, JavaScript tags, Tools: Google Analytics, Adobe Analytics, Matomo (overview), Data visualization and dashboard reporting, Applications: user tracking, campaign analysis, performance monitoring	8	L1,L2,L3
3	**SEO and User Behaviour Analysis** SEO fundamentals: on-page, off-page, and technical SEO, SEO metrics: organic traffic, bounce rate, backlinks, domain authority, Keyword research and content optimization, Tools: Google Search Console, Ahrefs, SEMrush, User segmentation: demographic, geographic, behavioural, Funnel analysis and user journey mapping, Applications: website visibility, engagement optimization	7	L1,L2,L3,L4
4	**Information Retrieval Models and Ranking Algorithms** IR models: Boolean, Vector Space, Probabilistic, Ranking algorithms: TF, IDF, TF-IDF, BM25, PageRank, HITS (Hyperlink-Induced Topic Search) Query processing: query formulation, expansion, Evaluation metrics: precision, recall, F1-score, MAP, Applications: e-commerce search, enterprise search	8	L1,L2,L3,L4
5	**Advanced Analytics and Recommender Systems** A/B testing and multivariate testing, Conversion Rate Optimization (CRO) Recommender systems: collaborative filtering, content-based filtering, Predictive analytics: intro to clustering and classification, Applications: personalization, marketing ROI, churn prediction	8	L2,L3,L4,L5
6	**Emerging Trends in Web and Information Analytics** Real-time analytics and stream processing, Semantic search and embeddings: Word2Vec, BERT, Social media and mobile analytics, Big data frameworks: Hadoop, Spark, Ethical issues: data privacy (GDPR, CCPA), consent management Applications: smart cities, IoT analytics, voice search	8	L1,L2,L3,L4,L5
Total Hours		**45**	

Capstone Project Guide Lines

1. The mini project work is to be conducted by a group of three students

2. Each group will be associated with a subject Incharge/ mini project mentor. The group should meet with the concerned faculty during Laboratory hours and the progress of work discussed must be documented.

3. Students may conduct a survey of different applications where Web Analytics and Information Retrieval techniques can be applied (e.g., e-commerce, blogging platforms, CRM, content recommendation systems).

4. Students will perform installation, configuration, and usage of latest Web Analytics and SEO tools (e.g., Google Analytics, Ahrefs, SEMrush, Google Search Console, Google Data Studio).

5. Students will try to consider following points in their Mini Project

a) Web data collection methods
b) Metric analysis (traffic, bounce rate, conversions)
c) Visualization of user behaviour and funnels
d) SEO audit and performance tracking
e) Keyword ranking and optimization strategy

6. Each group along with the concerned faculty shall identify a potential problem statement for project development, on which the study and implementation is to be conducted.

7. Each group may present their work in various project competitions and paper presentations.

8. A detailed report is to be prepared as per guidelines given by the concerned faculty.

Capstone Project Hours Distribution:

Sr. No.	Work to be done	No. of Hours	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Study WAIR tools, research papers, articles; identify mini project topics	4	L1, L2
2	Project title finalization and development of modules	2	L1, L2
3	Design methodology and choose tools (e.g., Google Analytics, Ahrefs)	4	L1, L2
4	Implementation of modules – Phase 1 (data collection, tracking setup)	4	L1, L2, L3
5	Result Phase I	2	L1, L2, L3, L4
6	Implementation of modules – Phase 2 (SEO audit, behaviour analysis)	4	L1, L2, L3
7	Result Phase II	2	L1, L2, L3, L4
8	Testing and metric validation (bounce rate, CTR, session analysis)	2	L1, L2, L3, L4
9	Result validation	2	L1, L2, L3, L4, L5
10	Report Writing	4	L1, L2
	Total Hours	**30**	

Books and References:

Sr. No.	Title	Authors	Publisher	Edition	Year
1	Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity	Avinash Kaushik	Sybex (Wiley)	1st Edition	2009
2	Digital Marketing Analytics: Making Sense of Consumer Data in a Digital World	Chuck Hemann, Ken Burbary	Que Publishing	2nd Edition	2018
3	Introduction to Information Retrieval	Christopher D. Manning, Prabhakar Raghavan, Hinrich Schütze	Cambridge University Press	1st Edition	2008
4	Search Engines: Information Retrieval in Practice	Bruce Croft, Donald Metzler, Trevor Strohman	Pearson	1st Edition	2010
5	SEO 2023: Learn Search Engine Optimization	Adam Clarke	Kindle/Independent	Latest Edition	2023
6	Google Analytics Breakthrough: From Zero to Business Impact	Feras Alhlou, Shiraz Asif, Eric Fettman	Wiley	1st Edition	2016
7	Practical Web Analytics for User Experience	Michael Beasley	Morgan Kaufmann	1st Edition	2013

Online References:

Sr. No.	Website Name	URL	Modules Covered
1	Google Analytics Academy	https://analytics.google.com/analytics/academy	M2, M3, M4
2	Ahrefs Blog & Academy	https://ahrefs.com/blog	M5, M6
3	SEMrush SEO Learning	https://www.semrush.com/academy/	M5
4	Google Search Central	https://developers.google.com/search	M5
5	Google Data Studio	https://lookerstudio.google.com	M4
6	Tableau Public	https://public.tableau.com	M4
7	Moz Beginner's Guide to SEO	https://moz.com/beginners-guide-to-seo	M5
8	Towards Data Science (Web Analytics)	https://towardsdatascience.com/tagged/web-analytics	M1, M3, M6
9	W3Schools Analytics & SEO	https://www.w3schools.com	M1, M2, M3, M5
10	HubSpot Marketing and Analytics Blog	https://blog.hubspot.com/marketing	M1, M2, M5, M6